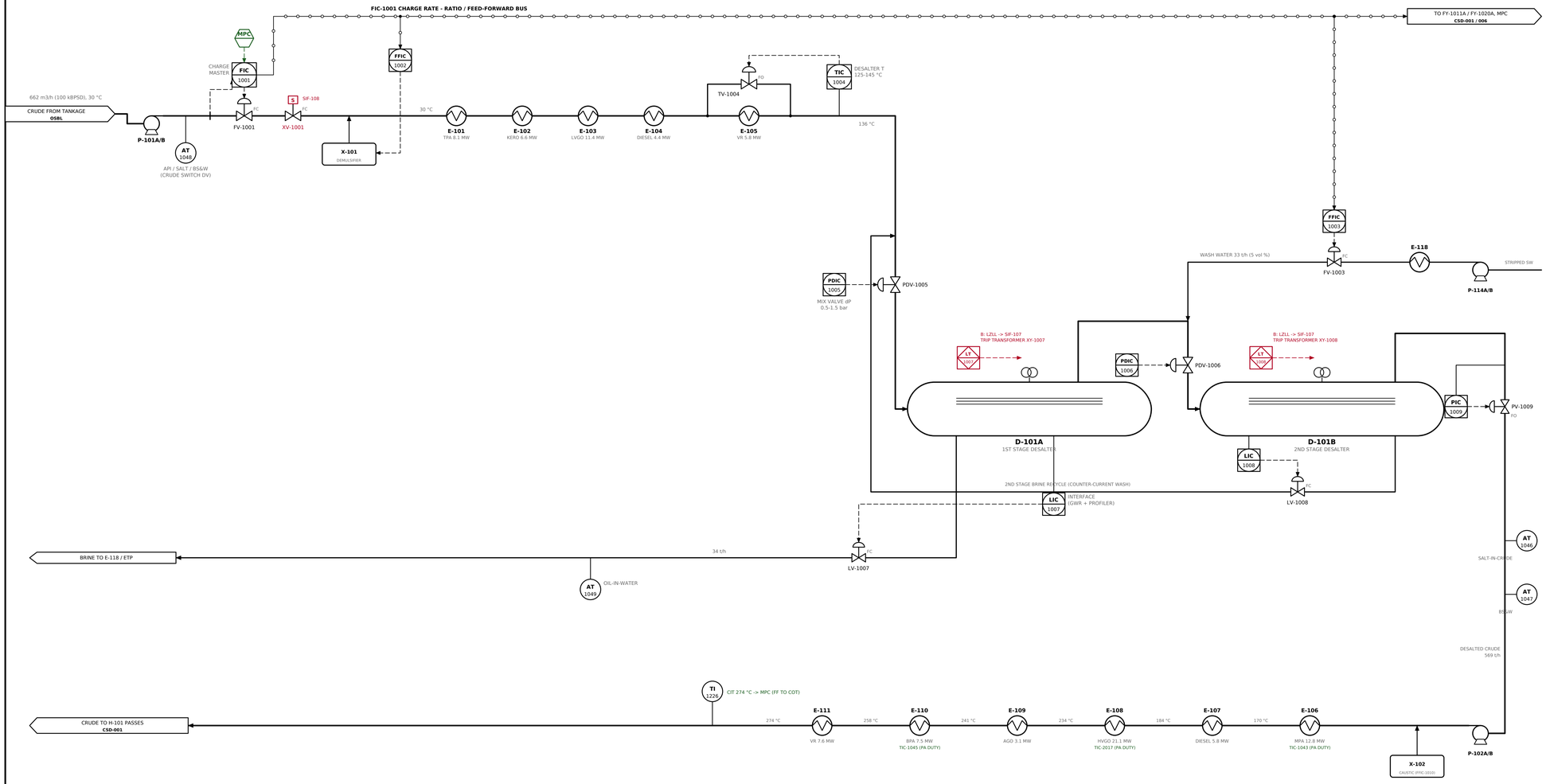


LEGEND (ISA-5.1 / ISA-5.4)

- DCS (BPCS) function, operator accessible
- SIS / BMS function (SIL rated)
- APC / MPC (L3 computer) function
- Field-mounted instrument
- Computing block: Σ sum/bias, × ratio/mult, < low select, > high select, f(x) characteriser
- Control valve (FO/FC = fail open/closed)
- Electrical signal (4-20 mA / HART)
- Software / data link (DCS internal)
- Process line
- APC setpoint / MPC move

CONTROL DESCRIPTION

1. FIC-1001 is the unit throughput master (operator or MPC SP). Its PV/SP is broadcast as ratio/feed-forward bus to demulsifier, wash water, caustic, heater passes (CSD-001) and the COT feed-forward.
2. Desalter temperature TIC-1004 (125-145 °C) positions the crude-side bypass of E-105; the VR side always flows. High temperature increases conductivity and grid current, low temperature impairs separation.
3. Mix-valve dP PDIC-1005/1006 set by operator 0.5-1.5 bar; optimised against salt-in-crude AT-1046 and oil-in-brine AT-1049 (no closed loop on analysers).
4. Interface LIC-1007/1008 (guided-wave radar with density profiler backup) - tight control; 2nd stage brine recycled counter-currently to 1st stage. Low-low interface (SIF-107) trips the transformers.
5. PIC-1009 holds the desalter pressure above crude vapour pressure at P-102 suction.
6. Hot-train exchanger bypasses are on the pumparound (hot) side and belong to PA duty control (CSD-002/005). CIT is a disturbance to the H-101 COT loop (feed-forward).
7. Crude switch: AT-1048 (density / salt) or tank-change signal triggers the MPC crude-switch feed-forward and the desalter chemical ratio presets.

PREHEAT & DESALTING - REGULATORY LOOPS

LOOP	SERVICE / ACTION	LIMITS
TIC-1004	Desalter inlet T via E-105 crude bypass TV-1004	125-145 °C
PDIC-1005/1006	Mixing-valve dP (emulsion / salt removal)	0.5-1.5 bar
LIC-1007/1008	Interface level, brine LV (tight, ±5 %)	LZLL SIF-107
PIC-1009	Desalter pressure > crude bubble point + 1.5 bar	10.0 barg
FFIC-1003	Wash water ratio to FIC-1001	4-6 vol %
FFIC-1002/1010	Demulsifier / caustic ratio to FIC-1001	ppm
TIC-1043/1045/2017	PA return T via exchanger PA-side bypass	MPC duty
FIC-1001	Unit throughput (MPC MV, max-feed push)	50-110 %

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)

GENERIC REFINERY - GULF COAST

100 KBPSD CRUDE & VACUUM DISTILLATION UNIT

PROPOSED CONTROL SCHEME

CRUDE CHARGE, DESALTERS & PREHEAT TEMPERATURE CONTROL

DISCIPLINE	STAGE	SCALE	SHEET
INSTRUMENTATION	FEED	NTS	3 OF 6

DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02

DRAWING NUMBER: **CFU-100-IC-CSD-003** — REV: **A**